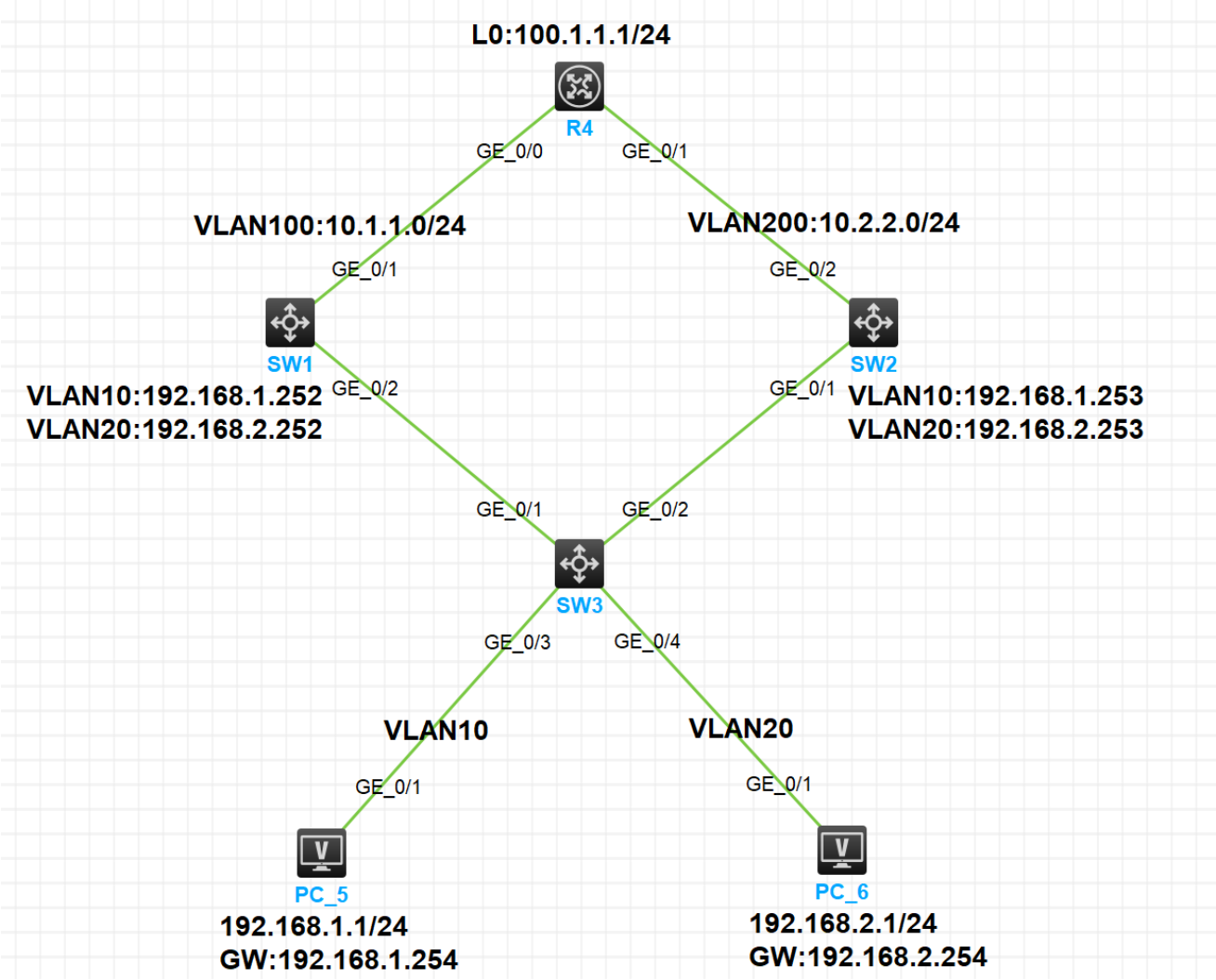


VRRP实验

实验拓扑

实验需求

- 1.按照图示配置 IP 地址和网关
- 2.在 SW1，SW2，SW3 上创建 Vlan10 和 Vlan20，对应 IP 网段如图，交换机之间链路允许所有 VLAN 通过
- 3.在 SW1 和 SW2 上配置 VRRP，要求 SW1 成为 Vlan10 的主网关，SW2 成为 Vlan20 的主网关；SW1 和 SW2 互为备份
- 4.SW1 和 SW2 对上行接口进行监视，如上行接口故障，会触发 VRRP 角色切换



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4. SW1 和 SW2 对上行接口进行监视，如上行接口故障，会触发 VRRP 角色切换

实验配置

```
[SW1]vlan 10
[SW1-vlan10]vlan 20
[SW1-vlan20]vlan 100
[SW1-vlan100]int vlan 100
[SW1-Vlan-interface100]ip ad 10.1.1.1 24
[SW1-Vlan-interface100]int vlan 10
[SW1-Vlan-interface10]ip ad 192.168.1.252 24
[SW1-Vlan-interface10]int vlan 20
[SW1-Vlan-interface20]ip ad 192.168.2.252 24
[SW1]int g1/0/1
[SW1-GigabitEthernet1/0/1]port link-type access
[SW1-GigabitEthernet1/0/1]port access vlan 100
[SW1-GigabitEthernet1/0/1]int g1/0/2
[SW1-GigabitEthernet1/0/2]port link-type trunk
[SW1-GigabitEthernet1/0/2]port trunk permit vlan 10 20
[SW1]int vlan 10
[SW1-Vlan-interface10]vrrp vrid 10 virtual-ip 192.168.1.254
[SW1-Vlan-interface10]vrrp vrid 10 priority 120
[SW1-Vlan-interface10]int vlan 20
[SW1-Vlan-interface20]vrrp vrid 20 virtual-ip 192.168.2.254
[SW1]track 1 interface g1/0/1
[SW1-track-1]int vlan 10
[SW1-Vlan-interface10]vrrp vrid 10 track 1 priority reduced 30

---
[SW2]vlan 10
[SW2-vlan10]vlan 20
[SW2-vlan20]vlan 200
[SW2-vlan200]int vlan 10
[SW2-Vlan-interface10]ip ad 192.168.1.253 24
[SW2-Vlan-interface10]int vlan 20
[SW2-Vlan-interface20]ip ad 192.168.2.253 24
[SW2-Vlan-interface20]int vlan 200
[SW2-Vlan-interface200]ip ad 10.2.2.2 24
[SW2-Vlan-interface200]int g1/0/2
[SW2-GigabitEthernet1/0/2]port link-type access
[SW2-GigabitEthernet1/0/2]port access vlan 200
[SW2-GigabitEthernet1/0/2]int g1/0/1
[SW2-GigabitEthernet1/0/1]port link-type trunk
[SW2-GigabitEthernet1/0/1]port trunk permit vlan 10 20
[SW2]int vlan 10
[SW2-Vlan-interface10]vrrp vrid 10 virtual-ip 192.168.1.254
[SW2-Vlan-interface10]int vlan 20
[SW2-Vlan-interface20]vrrp vrid 20 virtual-ip 192.168.2.254
[SW2-Vlan-interface20]vrrp vrid 20 priority 120
[SW2]track 1 interface g1/0/2
```

```
[SW2-track-1]int vlan 20
[SW2-Vlan-interface20]vrrp vrid 20 track 1 priority reduced 30
---
[SW3]vlan 10
[SW3-vlan10]vlan 20
[SW3]int ra g 1/0/1 to g 1/0/2
[SW3-if-range]port link-type trunk
[SW3-if-range]port trunk permit vlan 10 20
[SW3-if-range]int g1/0/3
[SW3-GigabitEthernet1/0/3]port link-type access
[SW3-GigabitEthernet1/0/3]port access vlan 10
[SW3-GigabitEthernet1/0/3]int g1/0/4
[SW3-GigabitEthernet1/0/4]port link-type access
[SW3-GigabitEthernet1/0/4]port access vlan 20
---
[R4]int g0/0
[R4-GigabitEthernet0/0]ip ad 10.1.1.4 24
[R4-GigabitEthernet0/0]int g0/1
[R4-GigabitEthernet0/1]ip ad 10.2.2.4 24
[R4-GigabitEthernet0/1]int l0
[R4-LoopBack0]ip ad 100.1.1.1 24
```

结果验证

1.接口正常时

```
[SW1]display vrrp
IPv4 virtual router information:
Running mode : Standard
Total number of virtual routers : 2
Interface          VRID  State      Running Adver    Auth    Virtual
                  pri    timer(cs) type      IP
-----
Vlan10             10   Master    120    100      None    192.168.1.254
Vlan20             20   Backup    100    100      None    192.168.2.254
<SW1>display vrrp verbose
IPv4 virtual router information:
Running mode : Standard
Total number of virtual routers : 2
Interface Vlan-interface10
VRID      : 10
Admin status : Up
Config pri : 120
Preempt mode : Yes
Auth type  : None
Virtual IP : 192.168.1.254
Virtual MAC : 0000-5e00-010a
Master IP  : 192.168.1.252
VRRP track information:
Track object : 1
State : Positive
Pri reduced : 30
```

```
Interface Vlan-interface20
  VRID          : 20          Adver timer : 100 centiseconds
  Admin status  : Up          State         : Backup
  Config pri    : 100         Running pri  : 100
  Preempt mode  : Yes         Delay time   : 0 centiseconds
  Become master : 3040 millisecond left
  Auth type     : None
  Virtual IP    : 192.168.2.254
  Master IP     : 192.168.2.253
```

```
[SW2]dis vrrp
```

IPv4 virtual router information:

Running mode : Standard

Total number of virtual routers : 2

Interface	VRID	State	Running pri	Adver timer(cs)	Auth type	Virtual IP
Vlan10	10	Backup	100	100	None	192.168.1.254
Vlan20	20	Master	120	100	None	192.168.2.254

<SW2>display vrrp verbose

IPv4 virtual router information:

Running mode : Standard

Total number of virtual routers : 2

Interface Vlan-interface10

```
  VRID          : 10          Adver timer : 100 centiseconds
  Admin status  : Up          State         : Backup
  Config pri    : 100         Running pri  : 100
  Preempt mode  : Yes         Delay time   : 0 centiseconds
  Become master : 2650 millisecond left
  Auth type     : None
  Virtual IP    : 192.168.1.254
  Master IP     : 192.168.1.252
```

Interface Vlan-interface20

```
  VRID          : 20          Adver timer : 100 centiseconds
  Admin status  : Up          State         : Master
  Config pri    : 120         Running pri  : 120
  Preempt mode  : Yes         Delay time   : 0 centiseconds
  Auth type     : None
  Virtual IP    : 192.168.2.254
  Virtual MAC   : 0000-5e00-0114
  Master IP     : 192.168.2.253
```

VRRP track information:

```
  Track object   : 1          State : Positive  Pri reduced : 30
```

2.将两台交换机上行接口全部关闭时

```
[SW1-GigabitEthernet1/0/1]shutdown
```

```
[SW2-GigabitEthernet1/0/2]shutdown
```

```

===
[SW1-GigabitEthernet1/0/1]display vrrp
IPv4 virtual router information:
Running mode : Standard
Total number of virtual routers : 2
Interface          VRID  State      Running Adver    Auth    Virtual
                   pri    timer(cs) type      IP
-----
Vlan10             10    Backup    90      100      None    192.168.1.254
Vlan20             20    Backup    100     100      None    192.168.2.254
---
[SW2-GigabitEthernet1/0/2]display vrrp
IPv4 virtual router information:
Running mode : Standard
Total number of virtual routers : 2
Interface          VRID  State      Running Adver    Auth    Virtual
                   pri    timer(cs) type      IP
-----
Vlan10             10    Master    100     100      None    192.168.1.254
Vlan20             20    Backup    90      100      None    192.168.2.254

```

3.只关闭SW1的上行接口

```

[SW1-GigabitEthernet1/0/1]display vrrp
IPv4 virtual router information:
Running mode : Standard
Total number of virtual routers : 2
Interface          VRID  State      Running Adver    Auth    Virtual
                   pri    timer(cs) type      IP
-----
Vlan10             10    Backup    90      100      None    192.168.1.254
Vlan20             20    Backup    100     100      None    192.168.2.254
---
<SW2>display vrrp
IPv4 virtual router information:
Running mode : Standard
Total number of virtual routers : 2
Interface          VRID  State      Running Adver    Auth    Virtual
                   pri    timer(cs) type      IP
-----
Vlan10             10    Master    100     100      None    192.168.1.254
Vlan20             20    Master    120     100      None    192.168.2.254

```

4.连通性测试

```

**PC5**
<H3C>ping 192.168.1.254
Ping 192.168.1.254 (192.168.1.254): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.1.254: icmp_seq=0 ttl=255 time=0.727 ms

```

```
56 bytes from 192.168.1.254: icmp_seq=1 ttl=255 time=0.808 ms

<H3C>ping 192.168.2.1
Ping 192.168.2.1 (192.168.2.1): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.2.1: icmp_seq=0 ttl=254 time=2.439 ms
56 bytes from 192.168.2.1: icmp_seq=1 ttl=254 time=1.324 ms

<H3C>ping 10.1.1.1
Ping 10.1.1.1 (10.1.1.1): 56 data bytes, press CTRL_C to break
56 bytes from 10.1.1.1: icmp_seq=0 ttl=255 time=0.398 ms
56 bytes from 10.1.1.1: icmp_seq=1 ttl=255 time=0.992 ms
---
**PC6**

<H3C>ping 192.168.2.254
Ping 192.168.2.254 (192.168.2.254): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.2.254: icmp_seq=0 ttl=255 time=1.192 ms
56 bytes from 192.168.2.254: icmp_seq=1 ttl=255 time=1.084 ms

<H3C>ping 192.168.1.1
Ping 192.168.1.1 (192.168.1.1): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.1.1: icmp_seq=0 ttl=254 time=1.154 ms
56 bytes from 192.168.1.1: icmp_seq=1 ttl=254 time=1.554 ms

<H3C>ping 10.2.2.2
Ping 10.2.2.2 (10.2.2.2): 56 data bytes, press CTRL_C to break
56 bytes from 10.2.2.2: icmp_seq=0 ttl=255 time=0.645 ms
56 bytes from 10.2.2.2: icmp_seq=1 ttl=255 time=1.011 ms
```

附录

1. 在实验配置完成后使用dis vrrp时，曾遇见过SW1、SW2的State均为Master，明显并不正确；

在仔细检查配置时，VRRP的配置均无问题；在系统日志报错时提示 "%Nov 17 18:44:43:789 2025 SW1 LLDP/5/LLDP_PVID_INCONSISTENT: PVID mismatch discovered on GigabitEthernet1/0/2 (PVID 1), with SW3 GigabitEthernet1/0/1 (PVID 10)."

解决：在报错中明显查看到说SW1的1/0/2端口和SW3的1/0/1端口的PVID错误，立即到该端口视图下使用dis th命令查看配置；发现是将端口设置为access端口并只放行了VLAN10，于是立即更改端口为trunk模式并放行了VLAN10、20

总结：端口的VLAN未放行或放行错误导致 VRRP 报文根本传不过去，所以两台交换机看各自都是Master；

在 SW1 上看：Vlan10 / Vlan20 都是 Master；在 SW2 上看：Vlan10 / Vlan20 也都是 Master。